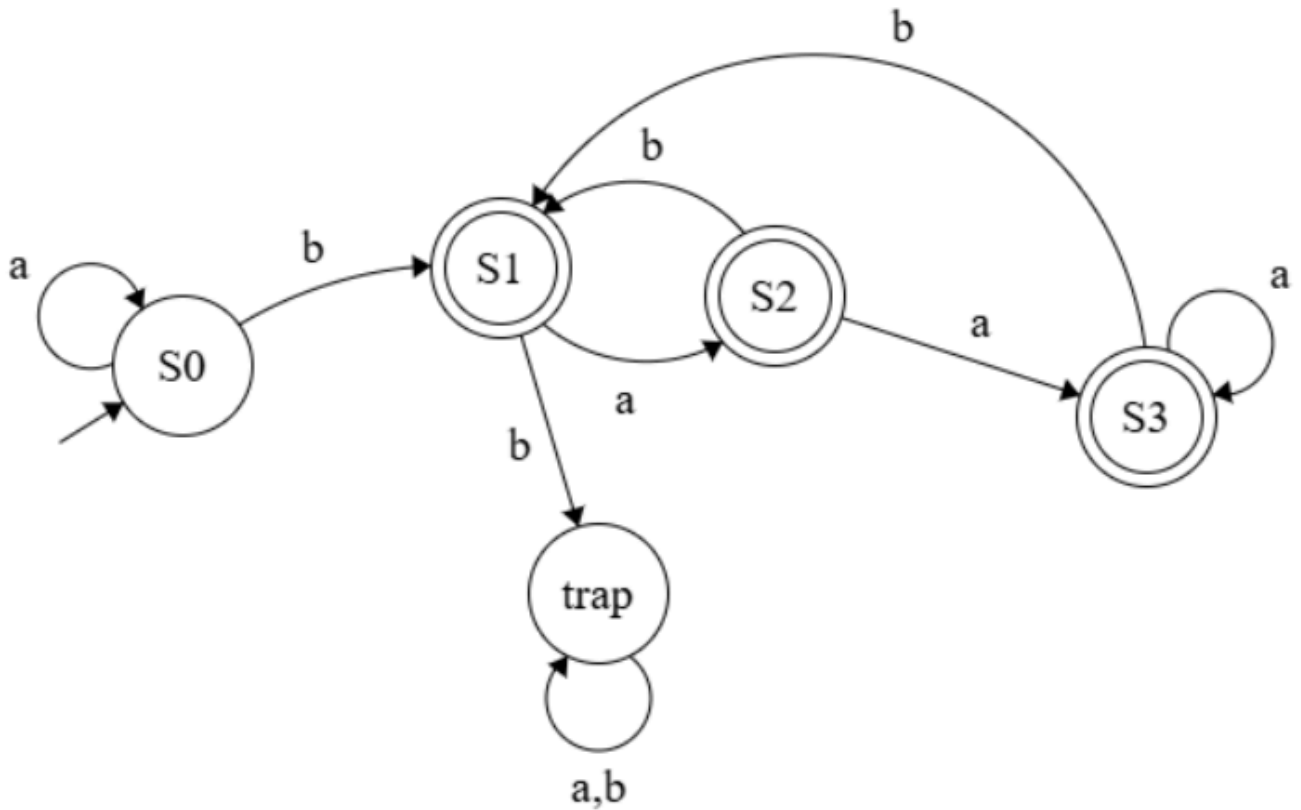


1A. Derive equivalent DFA M' to the provided NFA.



$S0 = \{A, B\}$

$S1 = \{C, D\}$

$S2 = \{B, C, D\}$

Trap = trap state

$S3 = \{A, B, C, D\}$

1B. Minimize M' and obtain the minimized DFA M''

Accepting states = $\{S1, S2, S3\}$ and non-accepting states = $\{S0, \text{trap}\}$

$Q' = \{\{S1, S2, S3\}, \{S0, \text{trap}\}\}$

Accepting Partition:

$S1$:

On $a = S2$

On $b = \text{trap}$

$S2$:

On $a = S3$

On $b = S1$

$S3$:

On $a = S3$

On $b = S1$

- S1 differs from S3, S2, therefore it differs and is distinguishable → split off from the accepting states.
- S2 and S3 on the other hand are indistinguishable → remain in the same state therefore.

Non-Accepting Partition:

S0:

On a = S0

On b = S1

Trap:

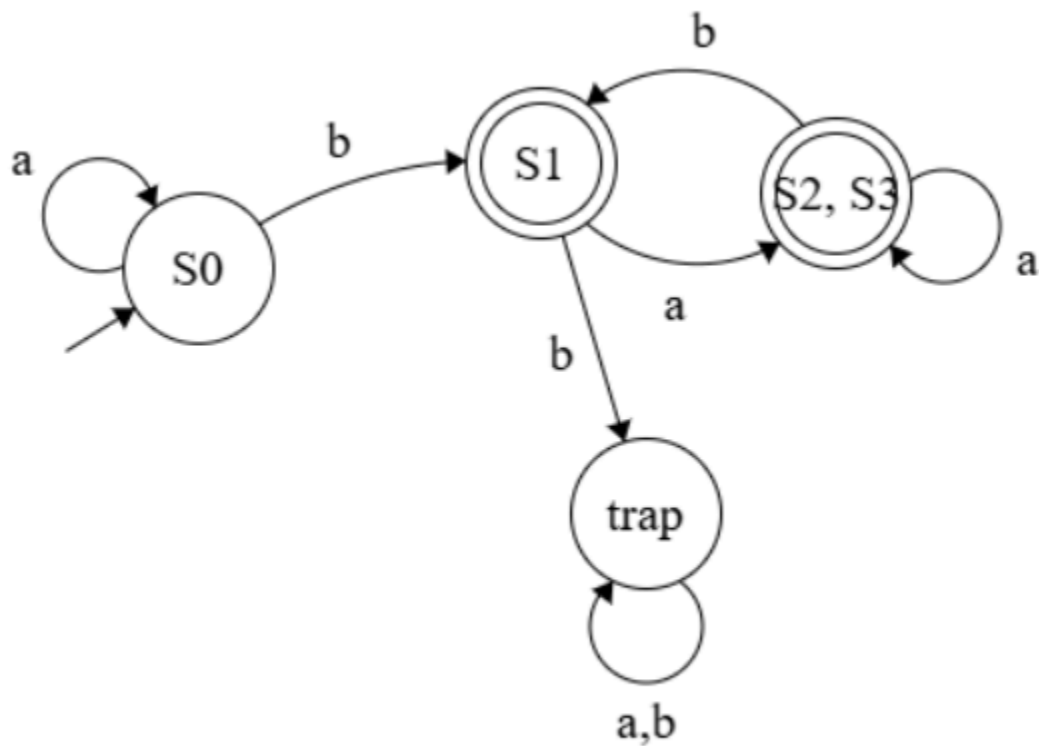
On a = trap

On b = trap

- S0 and trap differ and are therefore distinguishable

So now, $Q'' = \{\{S0\}, \{trap\}, \{S1\}, \{S2, S3\}\}$

This is the most minimized it can get. All necessary splits have been performed



1C. Describe in English the essential characteristics of the language accepted by the automata.

Strings in the automata: ab, abab, aba, abaaaaa

Strings not in automata: bb, aabb, aabbbbb

The automata's language is any combination of a's and b's where the substring 'bb' does not occur. Any combination of a's in b's in $\Sigma = \{a, b\}$ where a b is not consecutively repeated. If it is repeated, then the automata will not accept it and will go into trap state.

2. Prove that $L = \{ww : w \in \{0, 1\}^*\}$ is nonregular

Assume for contradiction that L is regular.

P is pumping length p that applies the three pumping lemma conditions to any string where string $s \in L$ with $|s| \geq p$

From L, $w = 0^p$ will be used so $s = ww \rightarrow 0^p * 0^p \rightarrow 0^{2p}$

$0^{2p} \in L$ and $|s| = 2p \geq p$

Split s into three parts xyz where $|xy| \leq p$ and $|y| > 0$

Since $|xy| \leq p$, x and y consist of only 0s from first half of s

$y = 0^k$ where $k \geq 1$

Pump y with $i = 2$

- This results in $s' = xy^2z$

This will then increase the 0s in the first half of the string by k:

- $s' = 0^{(p+k)}0^p = 0^{(p+k)}0^p$

$s' \in L$ if the first half equals the second half, but as observed above, the first half of s' is $0^{(p+k)}$ and the second half of s' is 0^p . These are not equal with $k \geq 1$, therefore $s' \notin L$

Therefore, there is a contradiction and L is not a regular language

3. Prove that $L = \{w\bar{w} : w \in \{0, 1\}^*\}$ is nonregular

Assume for contradiction that L is regular.

Pumping length p that for any string $s \in L$ with $|s| \geq p$

From L choose $w = 0^p$ so that $\bar{w} = 1^p$

So s is $w\bar{w} = 0^p1^p \mid s \in L$ and $|s| = 2p$ which is larger than p

$|xy| \leq p$ and $|y| > 0$, so both the x and y contain 0s from the first half of s

$y = 0^k$ ($k \geq 1$)

Pumping y with $i = 2$

- Results in $s' = xy^2z$

- Increases first half 0s by k $\rightarrow s' = 0^{(p+k)}1^p$

Does $s' \in L$?

- s' must consist of a string $w\bar{w}$ to belong to L $\rightarrow s' = w\bar{w}$

- In the pumped $s' = 0^{(p+k)}1^p$, the language rule does not apply and does not match the other strings in L.

- $0^{(p+k)}$ and 1^p are not bitwise complements as there are more 0s than 1s

Therefore, $s' \notin L$ as it contradicts the pumping lemma. This is a contradiction, and L is hence not a regular language.

4A. If L1 and L2 are nonregular, L1 U L2 must be nonregular.

Set L1 to the nonregular language $L1 = \{a^nb^n \mid 0 \leq n\}$

Set L2 to the nonregular language $L2 = \{b^na^n \mid 0 \leq n\}$

$L1 \cup L2 = \{a^nb^n \mid 0 \leq n\} \cup \{b^na^n \mid 0 \leq n\}$

For the sake of contradiction, assume that the union between the two languages is regular.

Use theorem that regular languages are closed under intersection with regular languages are regular

$R = a^*b^*$ is a regular language and chosen to be used to prove the contradiction.

$(L1 \cup L2) \cap R$

$L1 \subseteq a^*b^*$ as L1 consists of strings that have a's followed by b's

$L2 \cap a^*b^* = \emptyset$ because L2 consists of strings of b's before a's, which none are in a^*b^*

Therefore the union $(L1 \cup L2) \cap R = L1 \cap R = L1$ and therefore L1 is regular

But L1 is given to be nonregular, which presents a contradiction.

Hence, the assumption that $L1 \cup L2$ is regular must be false.

So, if L1 and L2 are nonregular, then $L1 \cup L2$ must be nonregular and the statement is proven true through contradiction.

4B. If L1 and L2 are nonregular, $L1 \cdot L2$ must be nonregular.

Assume for contradiction that $L1 \cdot L2$ is regular.

$L1 = \{a^nb^n \mid n \geq 0\}$

$L2 = \{a^mb^m \mid m \geq 0\}$

$L = L1 \cdot L2 = \{a^nb^na^mb^m \mid n, m \geq 0\}$

Add another language to the mix L3, which is regular and is $L3 = \{a^*b^*\}$

Use theorem that regular languages closed under intersection with regular languages are regular

$L \cap L3$

M must equal 0 for a's to be followed by b's in L, so it then becomes a^nb^n when $m = 0$

$L \cap L3 \rightarrow a^nb^n \cap a^*b^* = \{a^nb^n \mid n \geq 0\} = L1$

Therefore $L \cap L3 = L1$, but L1 is nonregular on its own, which presents a contradiction.

Hence, assumption that $L = L1 \cdot L2$ is regular when L1, L2 are nonregular has to be false.

Therefore, if L1 and L2 are nonregular, then $L1 \cdot L2$ has to be nonregular. The statement is proven true through contradiction.

$\subseteq \wedge \in \forall \exists \neg \Sigma \varepsilon \cup$